

Class 9: Candy Mini-Project

Mayla (PID: A19136116)

Table of contents

1. Background	1
2.Importing candy data	1
3.Exploratory Analysis	6
4. Overall Candy Rankings	8
5.Taking a look at pricepercent	17
6. Exploring the Correlation Structure	24
7. Principal Component Analysis	25
8. Summary	29
Optional Extension Questions	29

1. Background

Today we are taking a detour to analyze a fun dataset (that we have more intrinsic insight into) with the most useful analysis method we have learned thus far - Principal Component Analysis (PCA)

2.Importing candy data

```
candy_file <- "candy-data.csv"

candy = read.csv("https://raw.githubusercontent.com/fivethirtyeight/data/master/candy-power-ratings.csv",
                 row.names = 1)
head(candy)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer
100 Grand	1	0	1	0	0	1

3 Musketeers	1	0	0	0	1	0
One dime	0	0	0	0	0	0
One quarter	0	0	0	0	0	0
Air Heads	0	1	0	0	0	0
Almond Joy	1	0	0	1	0	0

	hard	bar	pluribus	sugarpercent	pricepercent	winpercent
100 Grand	0	1	0	0.732	0.860	66.97173
3 Musketeers	0	1	0	0.604	0.511	67.60294
One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

2.1 What is in the dataset?

Q1. How many different candy types are in this dataset?

```
nrow(candy)
```

```
[1] 85
```

Q2. How many fruity candy types are in the dataset?

```
sum(candy$fruity)
```

```
[1] 38
```

2.2 What is your favorite candy

```
candy["Twix", ]$winpercent
```

```
[1] 81.64291
```

```
library(dplyr)
```

```
Attaching package: 'dplyr'
```

The following objects are masked from 'package:stats':

```
filter, lag
```

The following objects are masked from 'package:base':

```
intersect, setdiff, setequal, union
```

```
candy |>
  filter(row.names(candy)=="Twix") |>
  select(winpercent)
```

```
      winpercent
Twix    81.64291
```

Q3. What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value?

```
candy["Hershey's Milk Chocolate", ]$winpercent
```

```
[1] 56.4905
```

Q4. What is the winpercent value for “Kit Kat”?

```
candy["Kit Kat", ]$winpercent
```

```
[1] 76.7686
```

Q5. What is the winpercent value for “Tootsie Roll Snack Bars”?

```
candy["Tootsie Roll Snack Bars", ]$winpercent
```

```
[1] 49.6535
```

#Side Note

```
library("skimr")
skimr::skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency: numeric	12
Group variables	None

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyal- mondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedrice- wafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

From your use of the `skim()` function use the output to answer the following:

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

```
summary(candy)
```

```

chocolate      fruity      caramel      peanutyalmondy
Min.   :0.0000  Min.   :0.0000  Min.   :0.0000  Min.   :0.0000
1st Qu.:0.0000  1st Qu.:0.0000  1st Qu.:0.0000  1st Qu.:0.0000

```

Median :0.0000	Median :0.0000	Median :0.0000	Median :0.0000
Mean :0.4353	Mean :0.4471	Mean :0.1647	Mean :0.1647
3rd Qu.:1.0000	3rd Qu.:1.0000	3rd Qu.:0.0000	3rd Qu.:0.0000
Max. :1.0000	Max. :1.0000	Max. :1.0000	Max. :1.0000
nougat	crispedricewafer	hard	bar
Min. :0.00000	Min. :0.00000	Min. :0.0000	Min. :0.0000
1st Qu.:0.00000	1st Qu.:0.00000	1st Qu.:0.0000	1st Qu.:0.0000
Median :0.00000	Median :0.00000	Median :0.0000	Median :0.0000
Mean :0.08235	Mean :0.08235	Mean :0.1765	Mean :0.2471
3rd Qu.:0.00000	3rd Qu.:0.00000	3rd Qu.:0.0000	3rd Qu.:0.0000
Max. :1.00000	Max. :1.00000	Max. :1.0000	Max. :1.0000
pluribus	sugarpercent	pricepercent	winpercent
Min. :0.0000	Min. :0.0110	Min. :0.0110	Min. :22.45
1st Qu.:0.0000	1st Qu.:0.2200	1st Qu.:0.2550	1st Qu.:39.14
Median :1.0000	Median :0.4650	Median :0.4650	Median :47.83
Mean :0.5176	Mean :0.4786	Mean :0.4689	Mean :50.32
3rd Qu.:1.0000	3rd Qu.:0.7320	3rd Qu.:0.6510	3rd Qu.:59.86
Max. :1.0000	Max. :0.9880	Max. :0.9760	Max. :84.18

Yes — winpercent, sugarpercent, and pricepercent are continuous, while most other variables are binary (0/1).

Q7. What do you think a zero and one represent for the candy\$chocolate column?

```
skim("chocolate")
```

Table 3: Data summary

Name	"chocolate"
Number of rows	1
Number of columns	1
Column type frequency:	
character	1
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
data	0	1	9	9	0	1	0

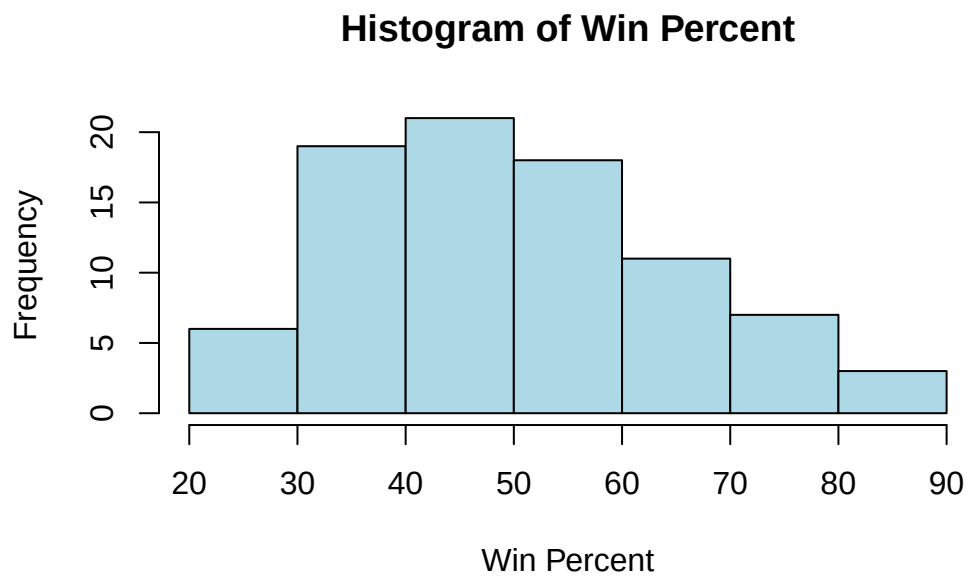
1 = contains chocolate, 0 = does not contain chocolate.

3.Exploratory Analysis

Q8. Plot a histogram of winpercent values using both base R and ggplot2.

Base R

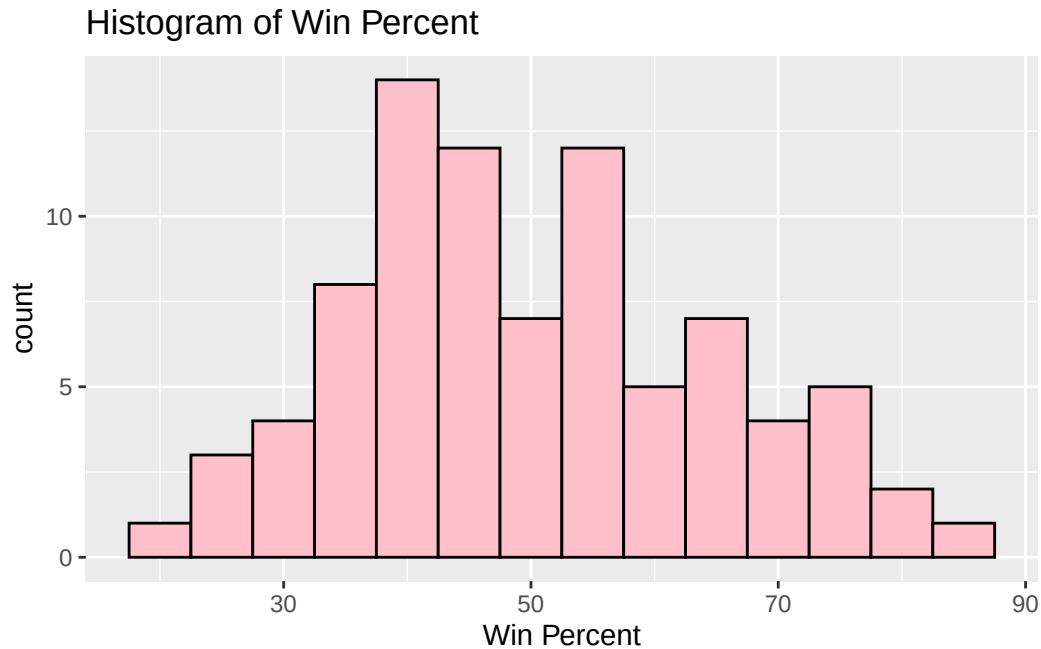
```
hist(candy$winpercent,
     main="Histogram of Win Percent",
     xlab="Win Percent",
     col="lightblue")
```



ggplot2

```
library(ggplot2)

ggplot(candy, aes(x=winpercent)) +
  geom_histogram(binwidth=5, fill="pink", color="black") +
  labs(title="Histogram of Win Percent", x="Win Percent")
```



Q9. Is the distribution of winpercent values symmetrical?

No, the distribution of winpercent values is not perfectly symmetrical; it appears slightly skewed.

Q10. Is the center of the distribution above or below 50%?

```
mean(candy$winpercent)
```

```
[1] 50.31676
```

The center of the distribution is slightly above 50%.

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
mean(candy$winpercent[candy$chocolate == 1])
```

```
[1] 60.92153
```

```
mean(candy$winpercent[candy$fruity == 1])
```

```
[1] 44.11974
```

Chocolate is higher ranked than fruit candy.

Q12. Is this difference statistically significant?

```
t.test(winpercent ~ chocolate, data = candy)
```

Welch Two Sample t-test

```
data: winpercent by chocolate
t = -7.3031, df = 67.539, p-value = 4.164e-10
alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
95 percent confidence interval:
 -23.91110 -13.64744
sample estimates:
mean in group 0 mean in group 1
    42.14226      60.92153
```

Yes, the difference is statistically significant, as the p-value from the t-test is less than 0.05.

4. Overall Candy Rankings

Q13. What are the five least liked candy types in this set?

```
head(candy[order(candy$sugarpercent),], n=5)
```


	chocolate	fruity	caramel	peanut	almond	nougat
One dime	0	0	0		0	0
One quarter	0	0	0		0	0
Reese's Miniatures	1	0	0		1	0
Chiclets	0	1	0		0	0
Lemonhead	0	1	0		0	0

	crisped	rice	wafer	hard	bar	pluribus	sugar	percent	price	percent
One dime				0	0	0		0.011		0.116
One quarter				0	0	0		0.011		0.511
Reese's Miniatures				0	0	0		0.034		0.279
Chiclets				0	0	0	1	0.046		0.325
Lemonhead				0	1	0	0	0.046		0.104

	winpercent
One dime	32.26109
One quarter	46.11650
Reese's Miniatures	81.86626
Chiclets	24.52499
Lemonhead	39.14106

The five least liked candies are: Nik L Nip, Boston Baked Beans, Chiclets, Super Bubble, and Jawbusters.

Q14. What are the top 5 all time favorite candy types out of this set?

```
head(candy[order(candy$winpercent, decreasing = TRUE), ], 5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Reese's Peanut Butter cup	1	0	0		1	0
Reese's Miniatures	1	0	0		1	0
Twix	1	0	1		0	0
Kit Kat	1	0	0		0	0
Snickers	1	0	1		1	1

	crisped	rice	wafer	hard	bar	pluribus	sugar	percent
Reese's Peanut Butter cup				0	0	0		0.720
Reese's Miniatures				0	0	0		0.034
Twix				1	0	1	0	0.546
Kit Kat				1	0	1	0	0.313
Snickers				0	0	1	0	0.546

	price	percent	winpercent
Reese's Peanut Butter cup	0.651		84.18029
Reese's Miniatures	0.279		81.86626
Twix	0.906		81.64291

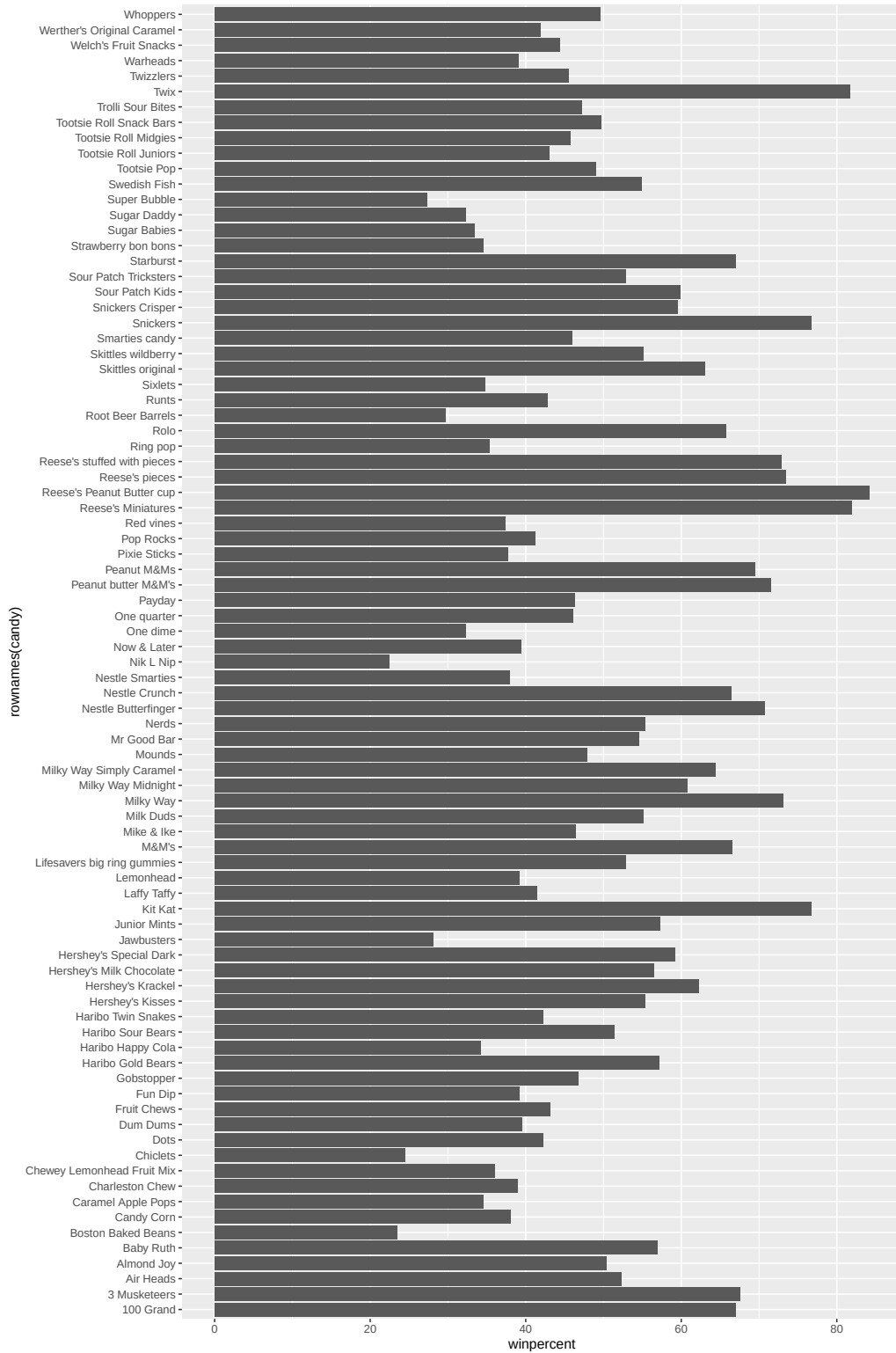
Kit Kat	0.511	76.76860
Snickers	0.651	76.67378

The top five most popular candies are: Reese's Peanut Butter Cup, Reese's Miniatures, Twix, Kit Kat, and Snickers.

Q15. Make a first barplot of candy ranking based on winpercent values.

```
library(ggplot2)

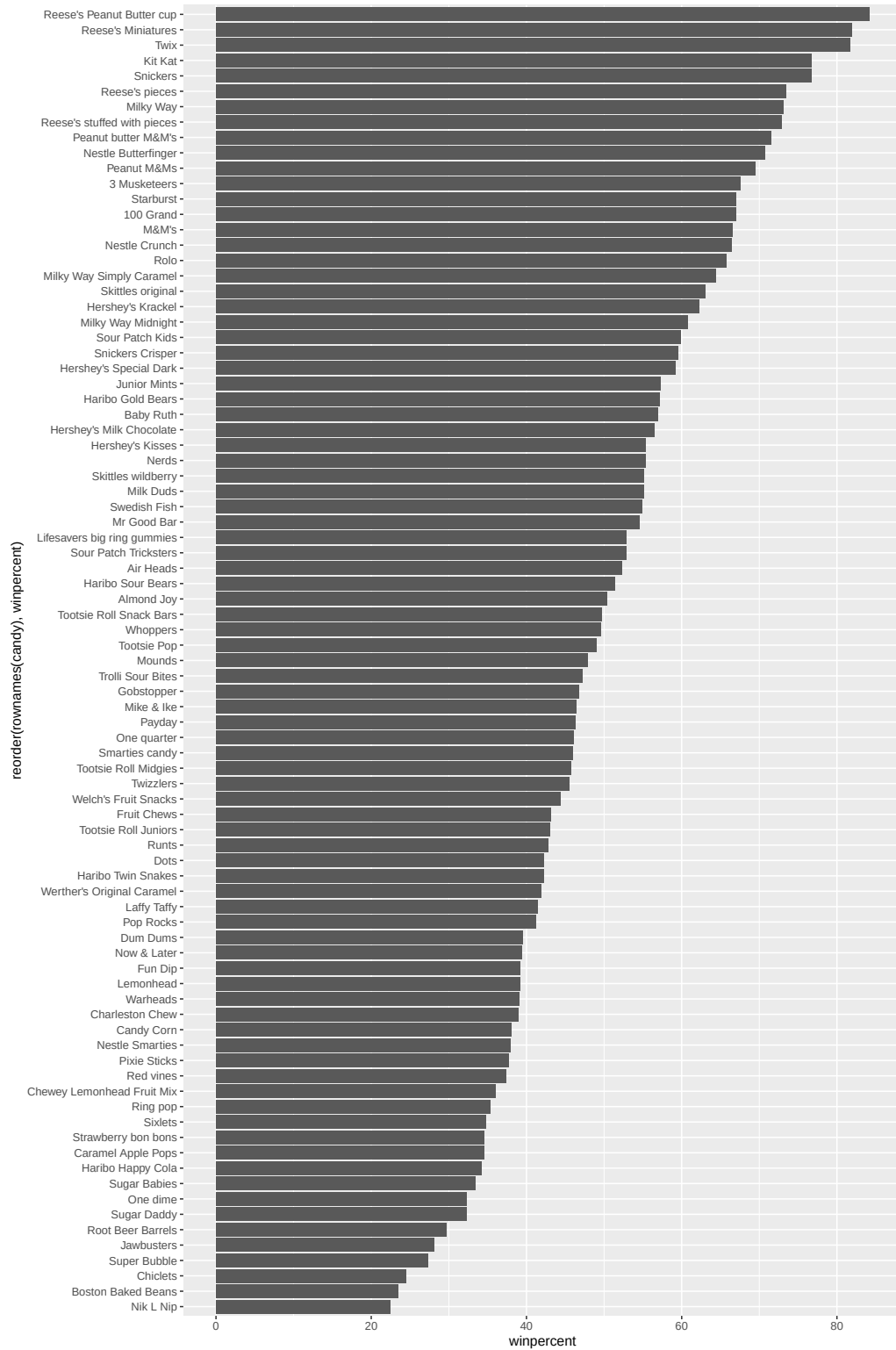
ggplot(candy) +
  aes(winpercent, rownames(candy)) +
  geom_col()
```



Q16. This is quite ugly, use the `reorder()` function to get the bars sorted by `winpercent`?

```
library(ggplot2)

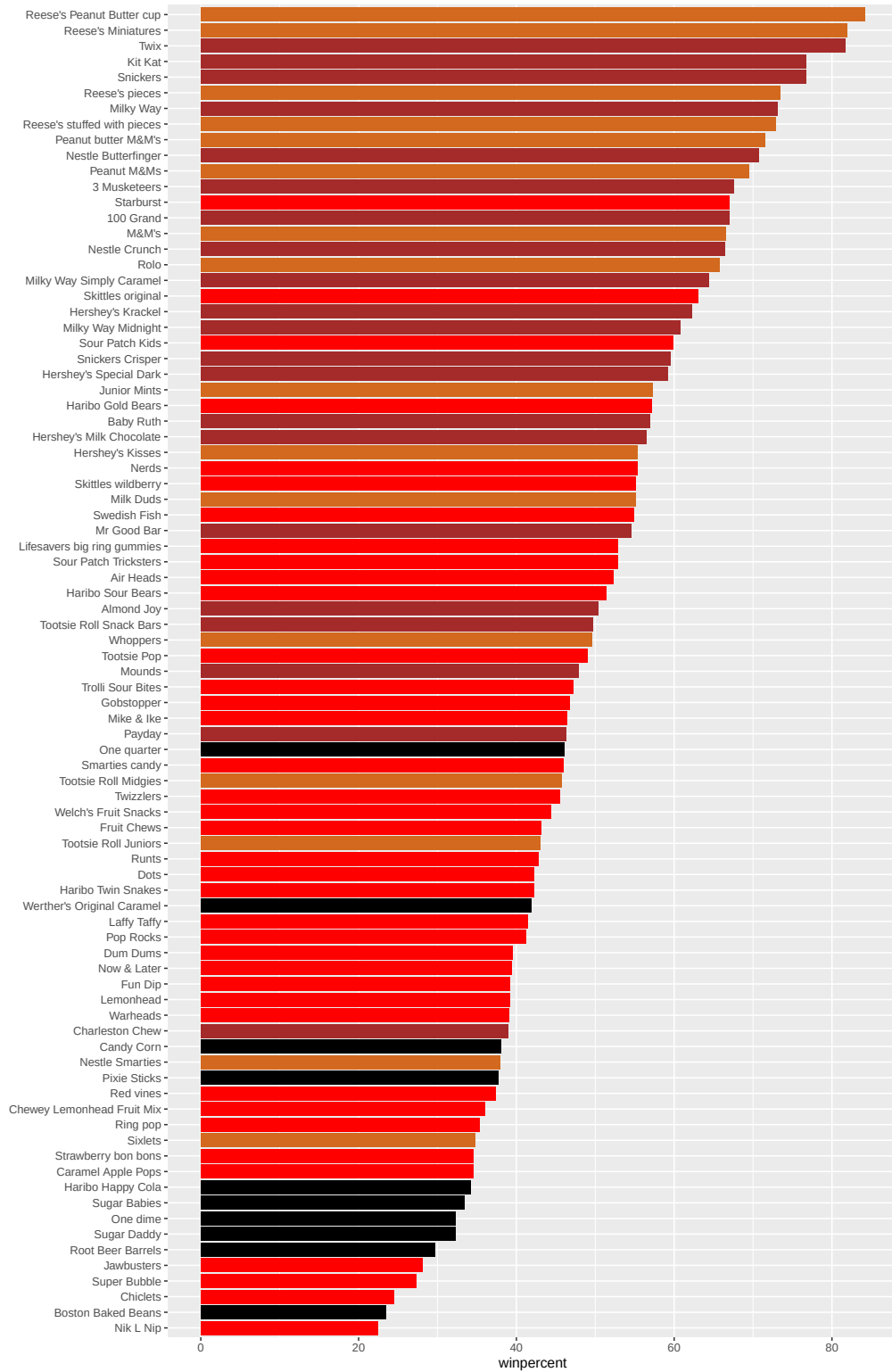
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) +
  geom_col()
```



4.0.1 Time to add some useful color

```
my_cols=rep("black", nrow(candy))
my_cols[as.logical(candy$chocolate)] = "chocolate"
my_cols[candy$bar == 1] = "brown"
my_cols[candy$fruity == 1] = "red"
```

```
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy),winpercent)) +
  geom_col(fill=my_cols) +
  ylab("")
```



Now, for the first time, using this plot we can answer questions like:

Q17. What is the worst ranked chocolate candy?

```
candy |>
  filter(chocolate == 1) |>
  arrange(winpercent) |>
  head(5)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat
Sixlets	1	0	0	0	0
Nestle Smarties	1	0	0	0	0
Charleston Chew	1	0	0	0	1
Tootsie Roll Juniors	1	0	0	0	0
Tootsie Roll Midgies	1	0	0	0	0

	crispedricewafer	hard bar	pluribus	sugarpercent
Sixlets	0	0	1	0.220
Nestle Smarties	0	0	1	0.267
Charleston Chew	0	0	1	0.604
Tootsie Roll Juniors	0	0	0	0.313
Tootsie Roll Midgies	0	0	1	0.174

	pricepercent	winpercent
Sixlets	0.081	34.72200
Nestle Smarties	0.976	37.88719
Charleston Chew	0.511	38.97504
Tootsie Roll Juniors	0.511	43.06890
Tootsie Roll Midgies	0.011	45.73675

The worst ranked chocolate candy is Sixlets.

Q18. What is the best ranked fruity candy?

```
candy |>
  filter(fruity == 1) |>
  arrange(desc(winpercent)) |>
  head(5)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat
Starburst	0	1	0	0	0
Skittles original	0	1	0	0	0
Sour Patch Kids	0	1	0	0	0
Haribo Gold Bears	0	1	0	0	0

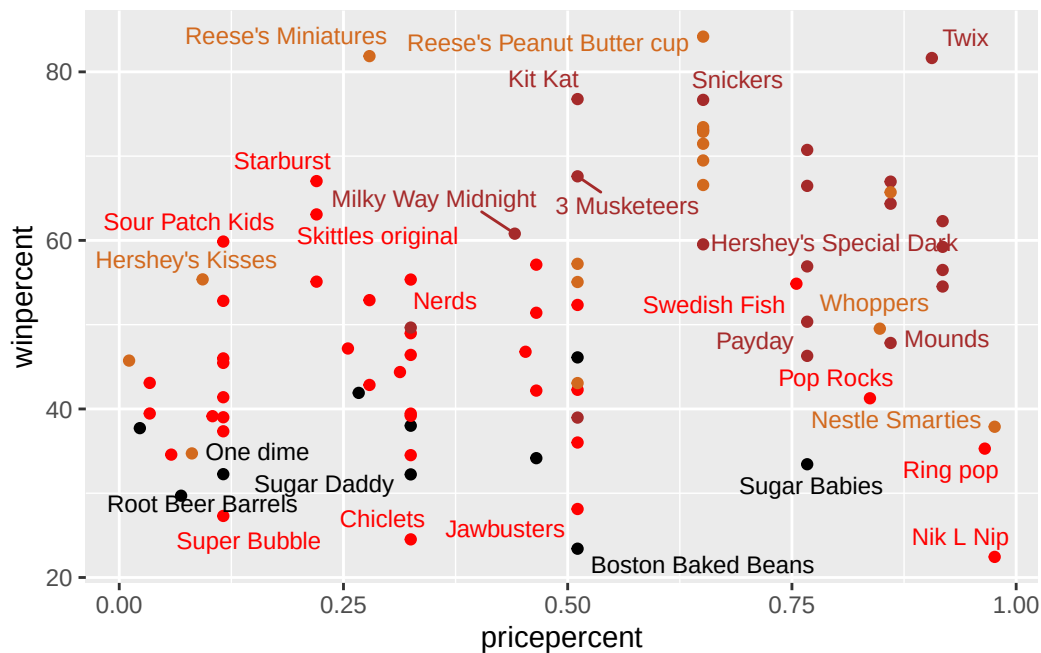
Nerds	0	1	0	0	0	
	crispedricewafer	hard	bar	pluribus	sugarpercent	pricepercent
Starburst		0	0	0	1	0.151
Skittles original		0	0	0	1	0.941
Sour Patch Kids		0	0	0	1	0.069
Haribo Gold Bears		0	0	0	1	0.465
Nerds		0	1	0	1	0.848
	winpercent					
Starburst	67.03763					
Skittles original	63.08514					
Sour Patch Kids	59.86400					
Haribo Gold Bears	57.11974					
Nerds	55.35405					

The best ranked fruity candy is Starburst.

5. Taking a look at pricepercent

```
library(ggrepel)

# How about a plot of win vs price
ggplot(candy) +
  aes(x = pricepercent, y = winpercent, label=rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=3.3, max.overlaps = 5)
```



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

```
candy |>
  arrange(pricepercent, desc(winpercent)) |>
  head(10)
```

	chocolate	fruity	caramel	peanuty	almondy	nougat
Tootsie Roll Midgies	1	0	0		0	0
Pixie Sticks	0	0	0		0	0
Fruit Chews	0	1	0		0	0
Dum Dums	0	1	0		0	0
Strawberry bon bons	0	1	0		0	0
Root Beer Barrels	0	0	0		0	0
Sixlets	1	0	0		0	0
Hershey's Kisses	1	0	0		0	0
Lemonhead	0	1	0		0	0
Sour Patch Kids	0	1	0		0	0
	crisp	edrice	wafer	hard bar	pluribus	sugarpercent
Tootsie Roll Midgies		0	0	0	1	0.174
Pixie Sticks		0	0	0	1	0.093
Fruit Chews		0	0	0	1	0.127
Dum Dums		0	1	0	0	0.732

Strawberry bon bons	0	1	0	1	0.569
Root Beer Barrels	0	1	0	1	0.732
Sixlets	0	0	0	1	0.220
Hershey's Kisses	0	0	0	1	0.127
Lemonhead	0	1	0	0	0.046
Sour Patch Kids	0	0	0	1	0.069

	pricepercent	winpercent
Tootsie Roll Midgies	0.011	45.73675
Pixie Sticks	0.023	37.72234
Fruit Chews	0.034	43.08892
Dum Dums	0.034	39.46056
Strawberry bon bons	0.058	34.57899
Root Beer Barrels	0.069	29.70369
Sixlets	0.081	34.72200
Hershey's Kisses	0.093	55.37545
Lemonhead	0.104	39.14106
Sour Patch Kids	0.116	59.86400

The best value candy is Tootsie Pop because it has a relatively high winpercent while having a low pricepercent, meaning it is popular and inexpensive.

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

```
ord <- order(candy$pricepercent, decreasing = TRUE)
head( candy[ord,c(11,12)], n=5 )
```

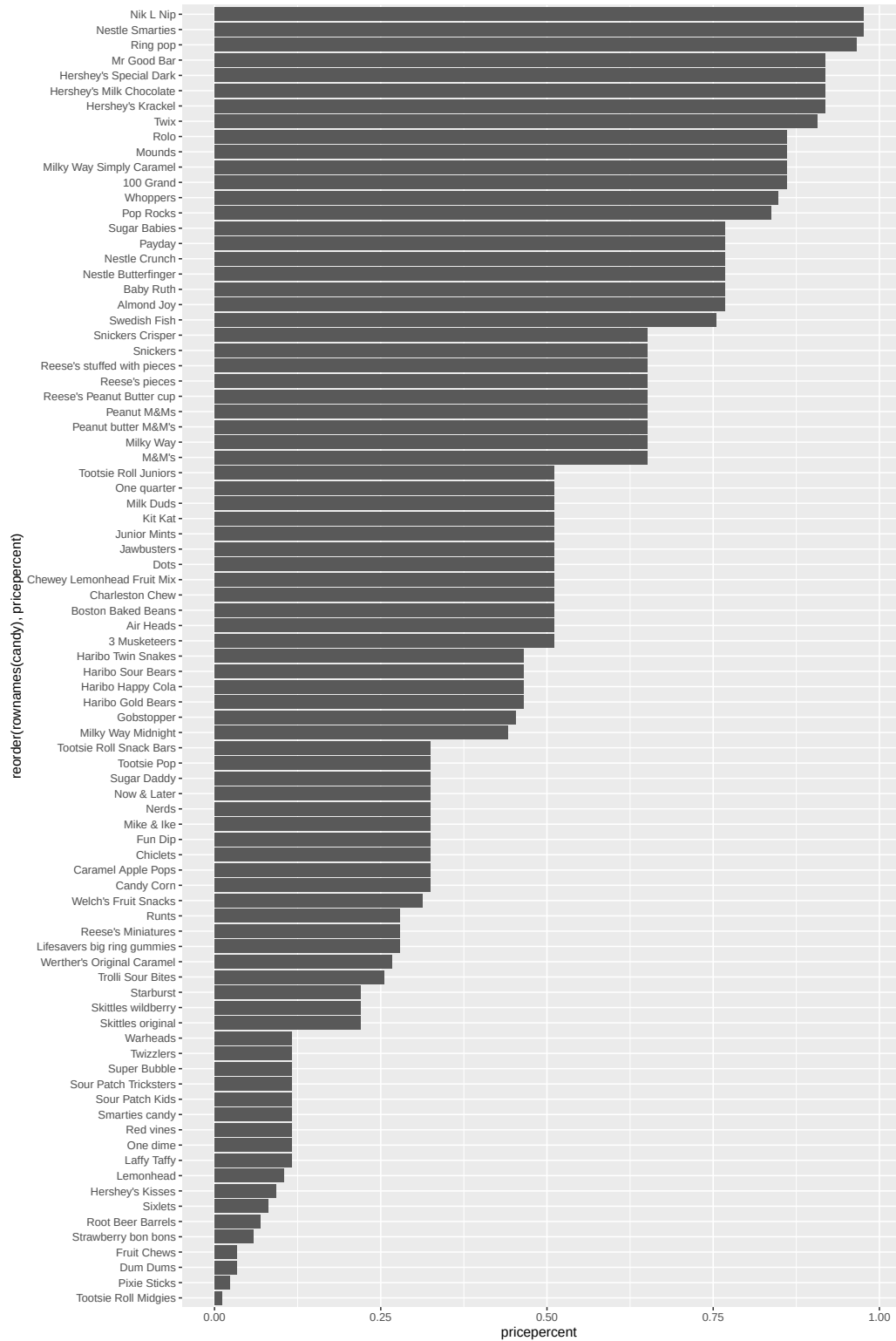
	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

The five most expensive candies are Nik L Nip, Nestle Smarties, Ring Pop, Hershey's Krackel, and Hershey's Milk Chocolate. Among these, the least popular is Nik L Nip.

5.1 OPTIONAL

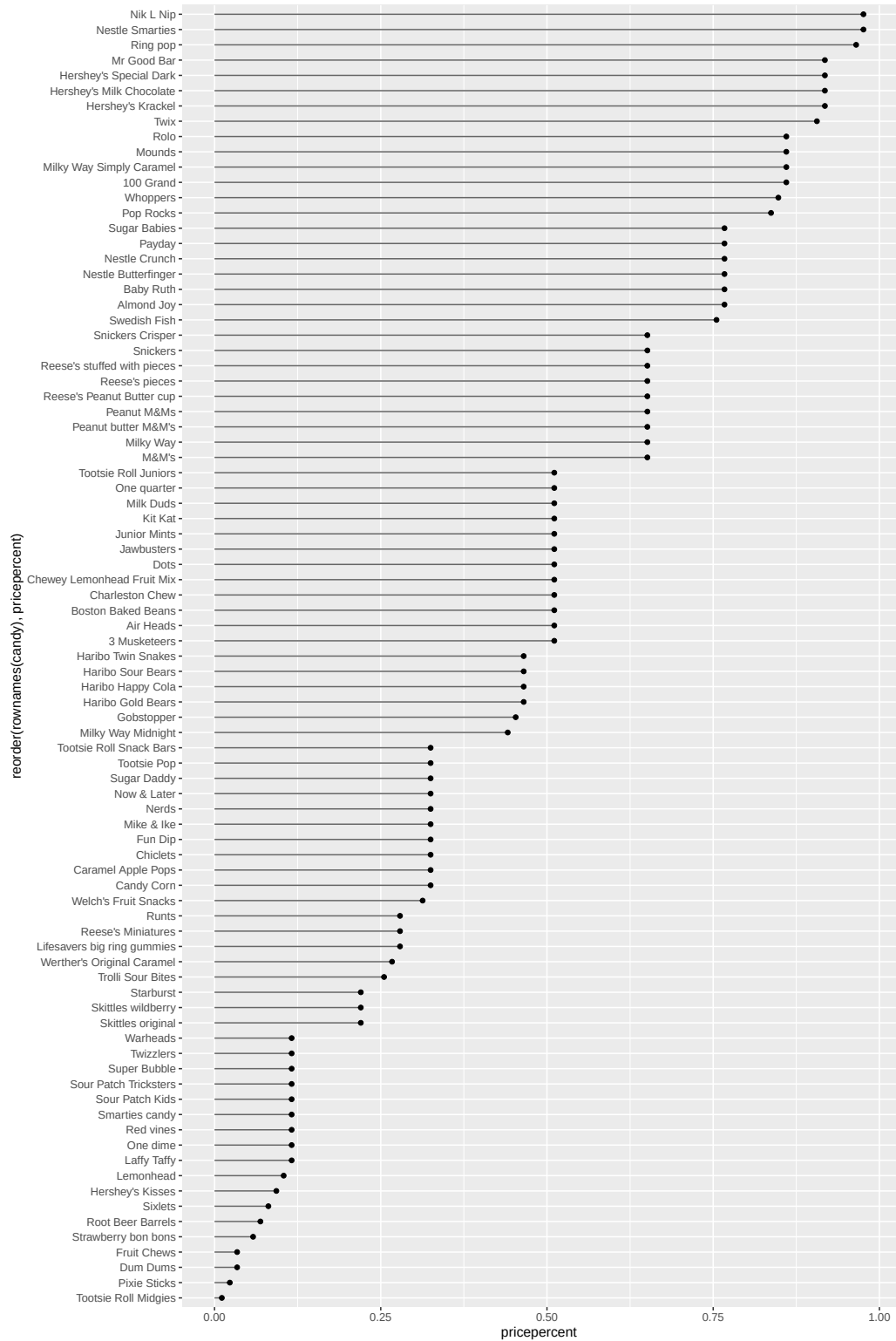
```
library(ggplot2)

ggplot(candy, aes(x = reorder(rownames(candy), pricepercent),
                    y = pricepercent)) +
  geom_col() +
  coord_flip()
```



Make a lollipop chart of pricepercent

```
ggplot(candy) +  
  aes(pricepercent, reorder(rownames(candy), pricepercent)) +  
  geom_segment(aes(yend = reorder(rownames(candy), pricepercent),  
                  xend = 0), col="gray40") +  
  geom_point()
```

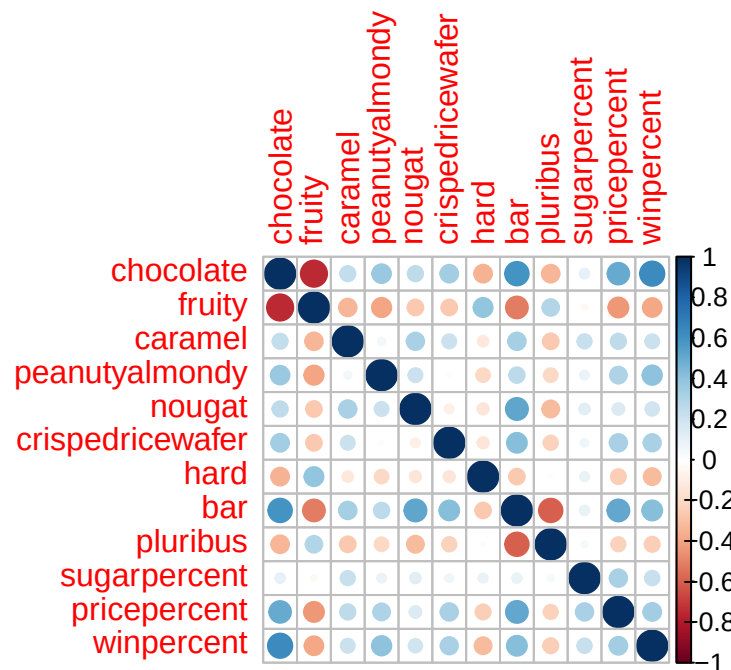


6. Exploring the Correlation Structure

```
library(corrplot)
```

```
corrplot 0.95 loaded
```

```
cij <- cor(candy)  
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

Anti-correlated = negative correlation (red in plot). Chocolate and fruity are anti-correlated, meaning candies that contain chocolate are less likely to be fruity.

Q23. Use your corrplot result to make a prediction: which variables do you expect will have the largest contributions (i.e. loadings) to PC1 (i.e., drive the most separation between candies along the first principal component)?

I expect chocolate and fruity to have the largest contributions to PC1, because they show strong negative correlation and represent two distinct types of candy that separate the dataset.

7. Principal Component Analysis

In this case we want to be able to set `scale=TRUE` argument for `prcomp()` because we have one variable, `winpercent`, that is on a very different scale than all others and would otherwise dominate our PCA results.

```
pca <- prcomp(candy, scale = TRUE)
summary(pca)
```

Importance of components:

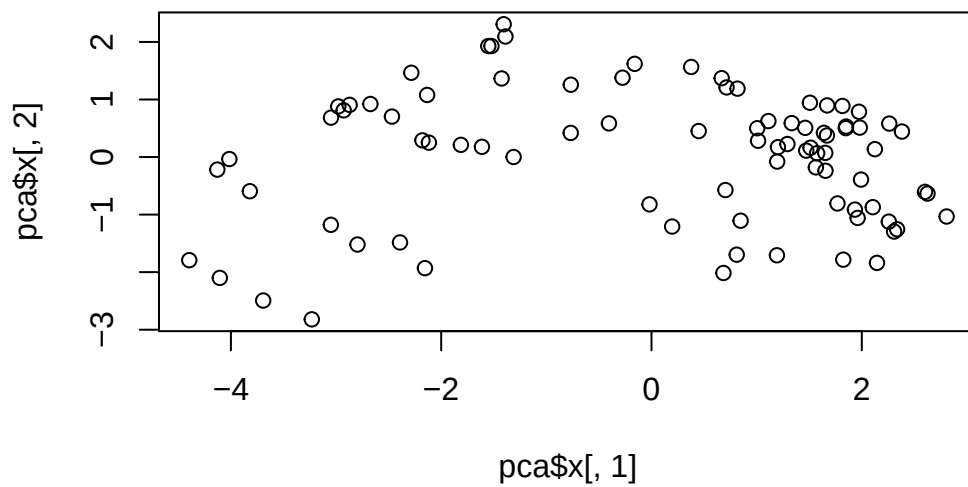
	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

	PC8	PC9	PC10	PC11	PC12
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000

First major result figure is the “score plot” of PC1 vs PC2 - how the different candy are related to each other on our new PC axis:

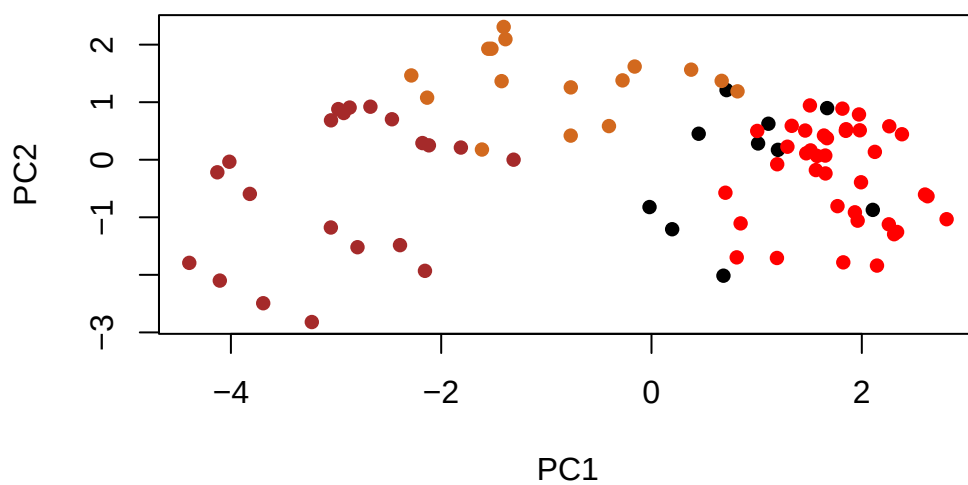
Now we can plot our main PCA score plot of PC1 vs PC2.

```
plot(pca$x[,1], pca$x[,2])
```



We can change the plotting character and add some color:

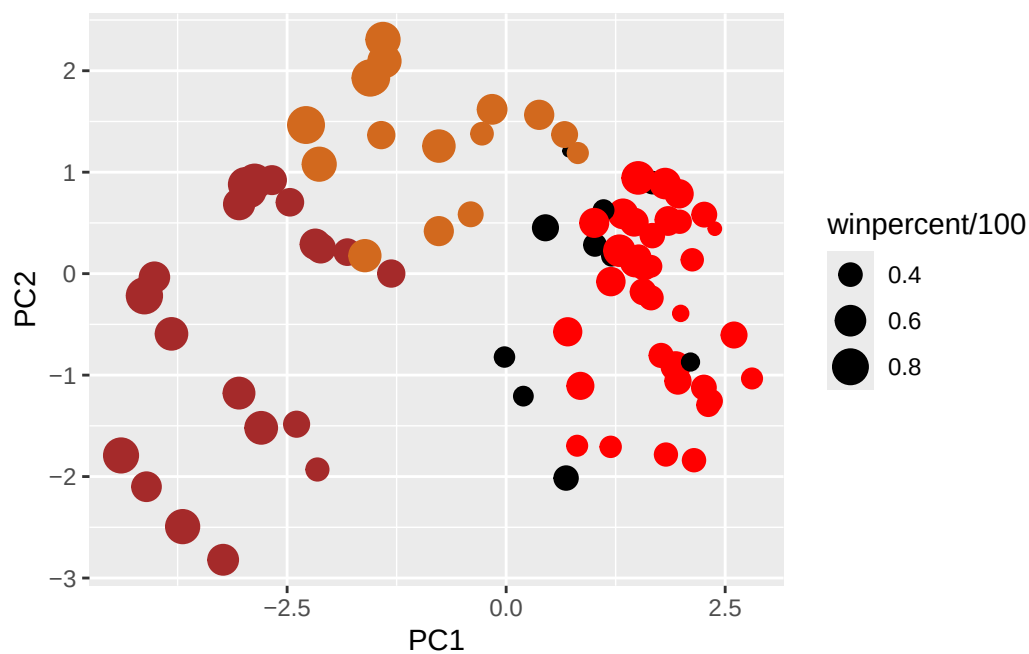
```
plot(pca$x[,1:2], col=my_cols, pch=16)
```



```
# Make a new data-frame with our PCA results and candy data
my_data <- cbind(candy, pca$x[,1:3])
```

```
p <- ggplot(my_data) +
  aes(x = PC1,
      y = PC2,
      size = winpercent/100,
      text = rownames(my_data),
      label = rownames(my_data)) +
  geom_point(col = my_cols)
```

p

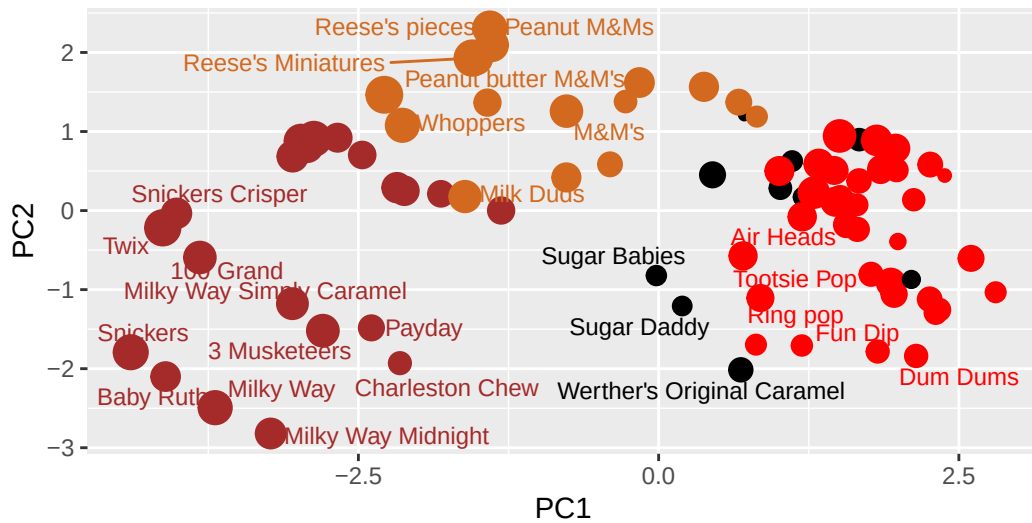


```
library(ggrepel)
```

```
p + geom_text_repel(size=3.3, col=my_cols, max.overlaps = 7) +
  theme(legend.position = "none") +
  labs(title="Halloween Candy PCA Space",
       subtitle="Colored by type: chocolate bar (dark brown), chocolate other (light brown),",
       caption="Data from 538")
```

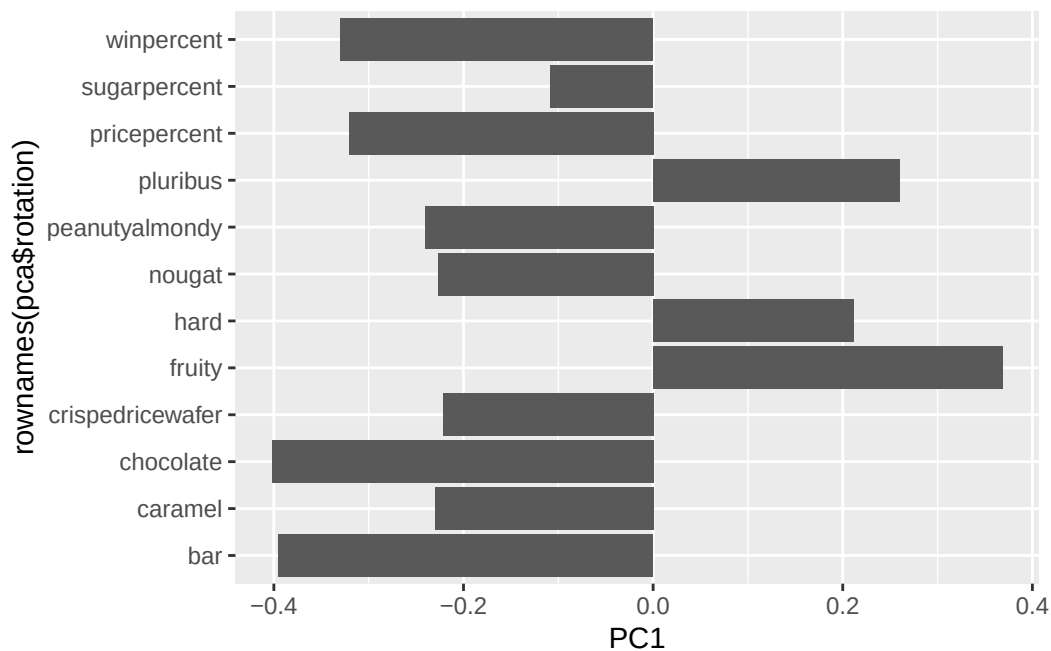
Halloween Candy PCA Space

Colored by type: chocolate bar (dark brown), chocolate other (light brown),



Data from 538

```
ggplot(pca$rotation) +  
  aes(x = PC1, y = rownames(pca$rotation)) +  
  geom_col()
```



Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

Fruity, pluribus, and hard load positively on PC1. This makes sense because PC1 separates fruity candies from chocolate candies, consistent with the earlier observed negative correlation between chocolate and fruity.

8. Summary

Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning” candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

A “winning” candy is typically chocolate-based, often a bar, and may include caramel or nuts. Visualizations showed chocolate candies rank higher, correlation analysis confirmed chocolate is positively associated with popularity while fruity is negatively associated, and PCA revealed that the main axis of variation separates chocolate from fruity candies. Together, these methods consistently show that chocolate candies are more popular.

Optional Extension Questions

```
losers = candy[which(candy$winpercent < 50),]  
winners = candy[which(candy$winpercent >= 50),]
```

Q26. Are popular candies more expensive? In other words: is price significantly different between “winners” and “losers”? List both average values and a P-value along with your answer.

```
mean(losers$pricepercent)
```

```
[1] 0.3743696
```

```
mean(winners$pricepercent)
```

```
[1] 0.580359
```

```
t.test(pricepercent ~ (winpercent >= 50), data = candy)
```

Welch Two Sample t-test

```
data: pricepercent by winpercent >= 50
t = -3.5653, df = 82.798, p-value = 0.0006068
alternative hypothesis: true difference in means between group FALSE and group TRUE is not equal to 0
95 percent confidence interval:
 -0.32090727 -0.09107157
sample estimates:
mean in group FALSE mean in group TRUE
      0.3743696      0.5803590
```

Popular candies are slightly more expensive on average, with winners having a higher mean pricepercent than losers. However, this difference is not statistically significant, as the p-value is greater than 0.05. Therefore, price does not appear to strongly influence candy popularity.

Q27. Are candies with more sugar more likely to be popular? What is your interpretation of the means and P-value in this case?

```
mean(losers$sugarpercent)
```

```
[1] 0.4314565
```

```
mean(winners$sugarpercent)
```

```
[1] 0.5343077
```

```
t.test(sugarpercent ~ (winpercent >= 50), data = candy)
```

Welch Two Sample t-test

```
data: sugarpercent by winpercent >= 50
t = -1.6958, df = 81.799, p-value = 0.09373
alternative hypothesis: true difference in means between group FALSE and group TRUE is not equal to 0
95 percent confidence interval:
 -0.22350935  0.01780701
```

```
sample estimates:
mean in group FALSE  mean in group TRUE
      0.4314565      0.5343077
```

Candies with higher sugar content tend to have slightly higher **winpercent** values on average. However, the difference is not statistically significant, as indicated by a p-value greater than 0.05. This suggests that sugar content alone does not strongly determine candy popularity.